OpenJLS

JPEG-LS Lossless Encoder IP Core — Datasheet

Version 1.0 • Single-component, fully pipelined, vendor-agnostic VHDL

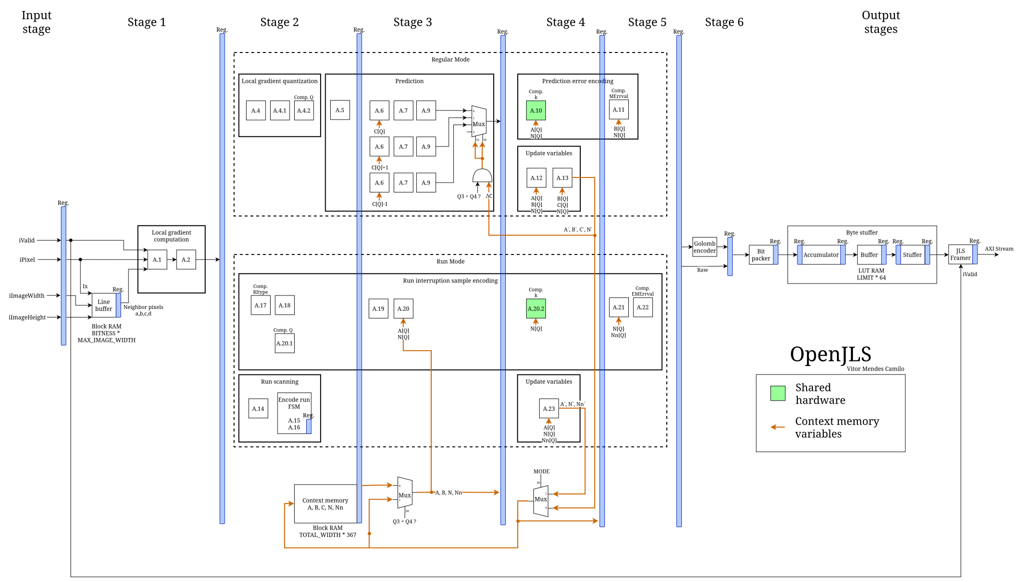
1 Overview

OpenJLS is a JPEG-LS (ISO/IEC 14495-1 / ITU-T T.87) lossless encoder for FPGAs. It encodes single-component (grayscale) images at one pixel per clock with no external memory, streaming a standards-compliant JPEG-LS bitstream out over AXI4-Stream. Source, license and the full verification suite are public at `https://github.com/VitorMendesC/OpenJLS`.

Standard	JPEG-LS lossless (ISO/IEC 14495-1, ITU-T T.87)
Components	Single (grayscale)
Pixel depth	8–16 bits (`BITNESS`)
Image size	Up to 65535 × 65535 px, minimum 4 × 1
Throughput	1 pixel / clock
Latency	14 clock cycles
f_{max}	∼240 MHz (Xilinx UltraScale+ `xczu7eg-1`, RTL-only)
Interface	AXI4-Stream input (pixels) and output (bitstream)
Memory	On-chip line buffer, one image row; no external RAM
Resources	∼8k LUT, ∼2.1k FF + BRAM scaling with image width
Portability	Vendor-agnostic VHDL (open-logic primitives)

2 Functional description

The core is a fully pipelined encoder with a latency of 14 clock cycles: an input register, six processing stages (gradient computation, context modeling, MED prediction with bias correction, Golomb–Rice / run-length coding) and three internally registered output stages (bit packer, byte stuffer, framer). End-of-image is derived internally from the configured dimensions; the output bitstream is self-delimiting.



3 Generics

Generic	Range	Description
`BITNESS`	8–16	Pixel bit depth.
`MAX_IMAGE_WIDTH`	4–65535	Largest width supported; sets line-buffer depth.
`MAX_IMAGE_HEIGHT`	1–65535	Largest height supported.
`OUT_WIDTH`	48–1024	Output bus width in bits; multiple of 8.

4 Ports

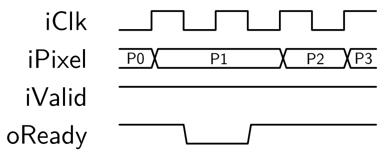
All ports are synchronous to the rising edge of `iClk`. The streaming ports map one-to-one onto AXI4-Stream as shown.

Port	Dir	Width	AXIS	Role
`iClk`	in	1	—	Clock.
`iRst`	in	1	—	Synchronous reset, active high; latches dimensions.
`iImageWidth`	in	$\lceil \log_2(\text{MAX\_W}+1) \rceil$	—	Image width, pixels (configuration).
`iImageHeight`	in	$\lceil \log_2(\text{MAX\_H}+1) \rceil$	—	Image height, pixels (configuration).
`iValid`	in	1	`TVALID`	Pixel valid.
`iPixel`	in	`BITNESS`	`TDATA`	Pixel data.
`oReady`	out	1	`TREADY`	Input ready.
`oData`	out	`OUT_WIDTH`	`TDATA`	Bitstream beat.
`oValid`	out	1	`TVALID`	Output valid.
`oKeep`	out	`OUT_WIDTH/8`	`TKEEP`	Valid-byte mask on final beat.
`oLast`	out	1	`TLAST`	End of image.
`iReady`	in	1	`TREADY`	Downstream ready.

The input omits `TLAST` (optional in AXI4-Stream); house-style port names map directly onto `s_axis`/`m_axis` for a thin wrapper.

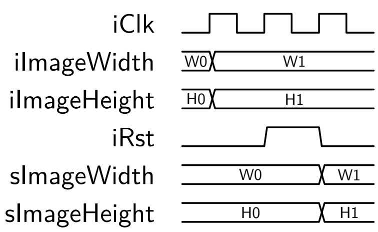
5 Timing and protocol

Pixel input. A pixel transfers on every cycle where `iValid` and `oReady` are both high. The core asserts `oReady` continuously and deasserts it only when the output FIFO fills under downstream backpressure.



`P2` is held flat across the stall and accepted when `oReady` returns high; no pixel is dropped while `oReady` is low.

Reset and configuration. The image dimensions are sampled while `iRst` is high: present `iImageWidth` and `iImageHeight` and hold them stable across a reset pulse; they latch into the internal registers on the next clock. Pulse reset again before streaming a new resolution. Tying the dimension ports to `0` selects the `MAX_IMAGE_*` maxima; an out-of-range value falls back to the maximum with a simulation warning.



The new size (`W1`/`H1`) is driven for several cycles, then `iRst` pulses high; the dimensions register one clock after the reset is sampled.

Output. Beats appear on `oData`/`oValid` subject to `iReady`; `oLast` marks the final beat and `oKeep` flags its valid bytes. Pipeline latency is 14 clock cycles.

6 Resources and performance

Characterized on Xilinx Zynq UltraScale+ `xczu7eg-fbvb900-1-e`, Vivado 2025.2, 12-bit, RTL-only (no floorplanning). f_{max} is true fmax (over-constrained until timing fails); at one pixel/clock ∼240 MHz is ∼240 Mpixel/s.

f_{max} **(MHz) vs image width**

Width	Default	Best-of†
4096	226.7	**249.2**
8192	241.0	**248.1**
12288	237.8	**250.4**
16384	201.5	**252.0**
32768	223.3	**242.0**
65535	235.0	**246.2**

Resources vs image width

Width	LUT	FF	BRAM
4096	7512	2055	1.5
8192	7545	2091	3.0
12288	7546	2123	4.5
16384	7674	2137	5.5
32768	7636	2137	11.0
65535	7781	2162	22.0

†Best of three implementation strategies (`Congestion_SpreadLogic_high`, `Performance_NetDelay_high`, `Performance_ExplorePostRoutePhysOpt`); the design is congestion-bound, so the winning strategy shifts with image size (results are deterministic, so the per-size winner is a reliable selection). Logic is essentially constant with image size; only the line-buffer BRAM scales with width and pixel depth.

7 Integration example

```
u_enc : entity work.openjls_top
  generic map (
    BITNESS => 8,
    MAX_IMAGE_WIDTH => 4096,
    MAX_IMAGE_HEIGHT => 4096,
    OUT_WIDTH => 64 )
  port map (
    iClk => clk,
    iRst => rst, -- hold high while dims are stable
    iImageWidth => img_w, -- latched on reset
    iImageHeight => img_h,
    iValid => px_valid, -- pixel stream (AXIS slave)
    iPixel => px_data,
    oReady => px_ready,
    oData => m_tdata, -- bitstream (AXIS master)
    oValid => m_tvalid,
    oKeep => m_tkeep,
    oLast => m_tlast,
    iReady => m_tready );
```

8 Conformance and verification

Verified by simulation with NVC using a layered, self-checking suite run at both RTL and post-synthesis (gate-level) stages:

- **OSVVM** — per-module tests against independent T.87-derived reference models, plus top-level control-plane stress (reset injection, backpressure, randomized sizes) with requirements tracking.
- **Coverage** — OSVVM functional coverage with NVC structural code coverage (99%+ statements).
- **Golden model** — byte-exact vs the independent CharLS C++ encoder over 287 real and synthetic images, plus ISO/IEC 14495-1 vectors.
- **Design contracts** — embedded PSL assertions (AXI4-Stream protocol, internal handshakes) on every run.
- **Post-synthesis** — the stress test and a golden-model subset re-run on the synthesized netlist.

Latest regression snapshot (live, per-test reports at `https://vitormendesc.github.io/OpenJLS/`):

Suite	Status	Test/Cov	Summary
OSVVM suite	PASS	100%	36 tests, 129 810 affirmations (module + top control-plane)
NVC code coverage	info	99.2%	Per-file statement breakdown
Golden model	PASS	100%	287/287 images byte-exact vs CharLS
Post-synth OSVVM	PASS	100%	Control-plane stress on the gate-level netlist
Post-synth golden model	PASS	100%	156/156 images byte-exact vs CharLS

Revision history

Version	**Date**	**Notes**
1.0	2026-06	Initial release.

References. ITU-T T.87 / ISO/IEC 14495-1; Y. M. Mert, *Key Architectural Optimizations for Hardware Efficient JPEG-LS Encoder*, IEEE 2018; CharLS; open-logic.